

**4** Movement in mammals results from interactions between muscles, tendons, ligaments and bones.

(a) How many of the following statements about tendons are correct?

(1)

- tendons act as extensors
- tendons are myogenic
- tendons connect muscles to bones

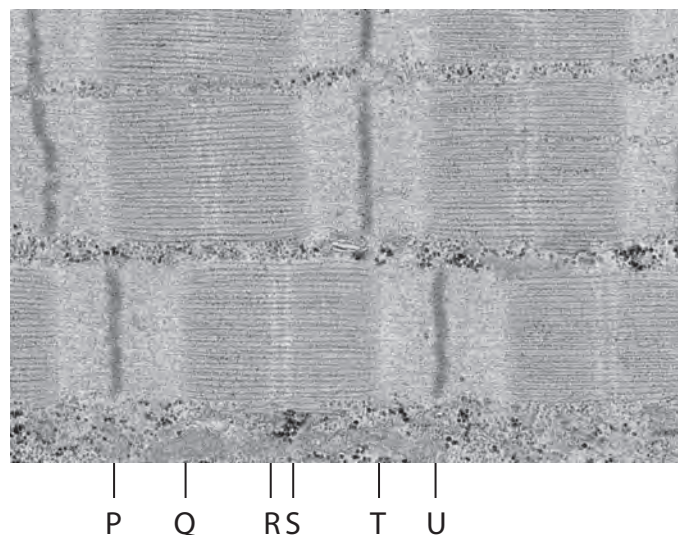
☐ **A** 0

☐ **B** 1

☐ **C** 2

☐ **D** 3

(b) The photograph shows part of a skeletal muscle fibre as seen using an electron microscope.

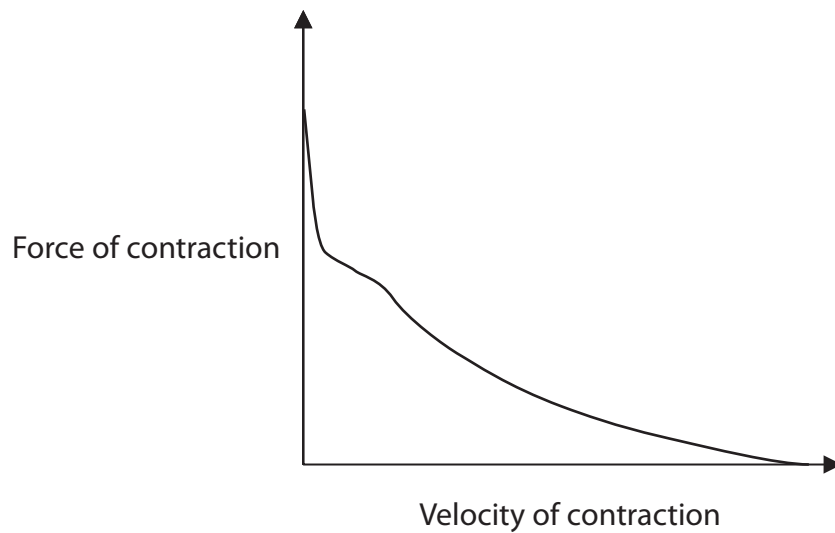


Scale  
1 µm

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- (c) The graph shows the relationship between the force of muscle contraction and the velocity of muscle contraction.



The greater the number of cross bridges formed between actin and myosin the greater the force of contraction of a muscle.

Explain the relationship between the force and the velocity of contraction of a muscle.

(3)

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(Total for Question 4 = 11 marks)